

Tutorial 2

Q1.

A. Calculate I_o
 $I_o = -6 \text{ mA}$

B. Nodes = 3, $B = 6$, $M = 6 - 3 + 1 = 4$

Nodal analysis:-

→ Currents are in mA

→ Resistance in $k\Omega$

$$-3I_1 - V_1 = 0$$

$$I_1 = -\frac{V_1}{3}$$

$$V_1 - V_2 = 2(I_1 - 6) \quad \text{--- (A)}$$

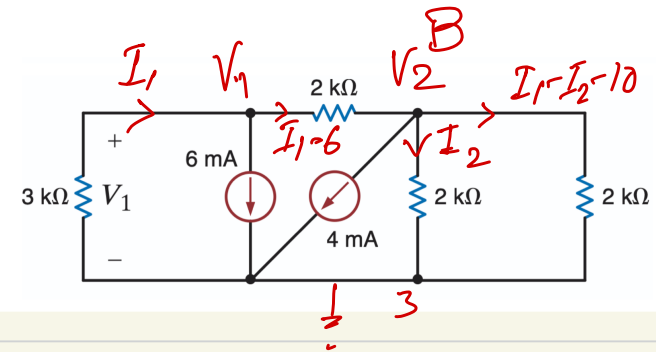
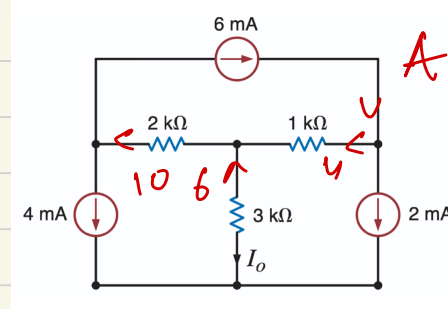
$$2I_2 = 2(I_1 - I_2 - 10) \quad \text{--- (B)}$$

$$V_2 = 2I_2 \Rightarrow I_2 = \frac{V_2}{2}$$

$$\text{(A)} \rightarrow V_1 - V_2 = -\frac{2V_1}{3} - 12 \Rightarrow \frac{5V_1}{3} - V_2 = -12$$

$$\text{(B)} \rightarrow V_2 = -\frac{2V_1}{3} - V_2 - 20 \Rightarrow \frac{2V_1}{3} + 2V_2 = -20$$

$$2\text{(A)} + \text{(B)} \rightarrow \frac{10V_1}{3} + \frac{2V_1}{3} = -24 - 20 \Rightarrow 4V_1 = -44 \Rightarrow \underline{V_1 = -11 \text{ V}}$$



Q.2 Dependent sources in the circuit.
Find V_1, V_2, V_3, V_4 .

Nodes: 5, also non-planar circuit.

Therefore, let's go ahead with Nodal Analysis.

$$V_2 - V_1 = 3I_x \quad \text{--- (A)}$$

$$V_4 - V_2 = 6I_x \quad \text{--- (B)}$$

$$V_3 = 12V \quad \text{--- (C)}$$

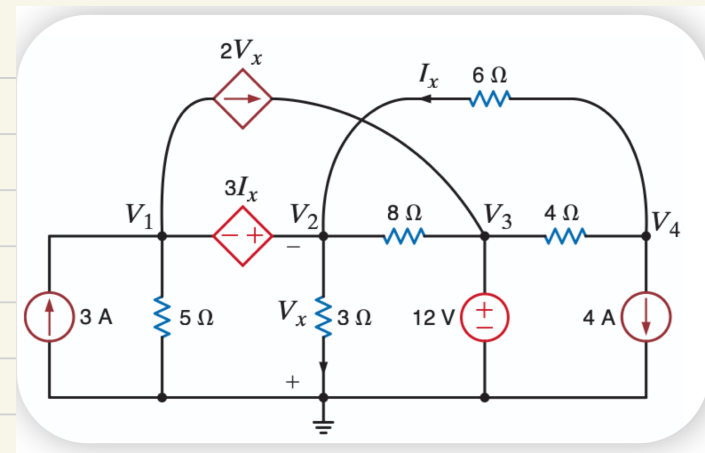
In this circuit, voltage across current sources are unknown. Similarly, current passing through voltage sources are unknown which makes it difficult to apply KCL at those nodes.

Using the Concept of supernode :- (V_1 & V_2)

$$V_x = -V_2 \quad \text{--- (D)}$$

re writing (1) with the help of (A), (C), (D) $\Rightarrow$

$$\begin{aligned} & -3 + \frac{V_1}{5} + 2V_x + \frac{V_2}{3} + \frac{V_2 - V_3}{8} - I_x = 0 \quad \text{--- (1)} \\ & \Rightarrow \frac{8}{15}V_1 - \frac{15}{8}V_2 - \frac{36}{8} = 0 \quad \text{--- (2)} \end{aligned}$$



at node 4

$$I_x + 4 + \frac{V_4 - V_3}{4} = 0$$

using (B) & (C) $\rightarrow$

$$I_x + 4 + \frac{6I_x + V_2 - 3}{4} = 0$$

$$\frac{5I_x}{2} + \frac{V_2}{4} + 1 = 0$$

using (A) $\rightarrow$

$$\frac{5}{6}(V_2 - V_1) + \frac{V_2}{4} + 1 = 0$$

$$\Rightarrow -\frac{5}{6}V_1 + \frac{13}{12}V_2 + 1 = 0 \quad \text{--- (3)}$$

$$(2) \times \frac{5}{6} + (3) \times \frac{8}{15} \Rightarrow \left(-\frac{5}{6} \times \frac{15}{8} + \frac{8}{15} \times \frac{13}{12}\right)V_1 - \frac{36}{8} \times \frac{5}{6} + \frac{8}{15} = 0$$

$$\Rightarrow V_2 = -3.267 \text{ V}$$

$$\text{using (3)} \rightarrow V_1 = -3.047 \text{ V}$$

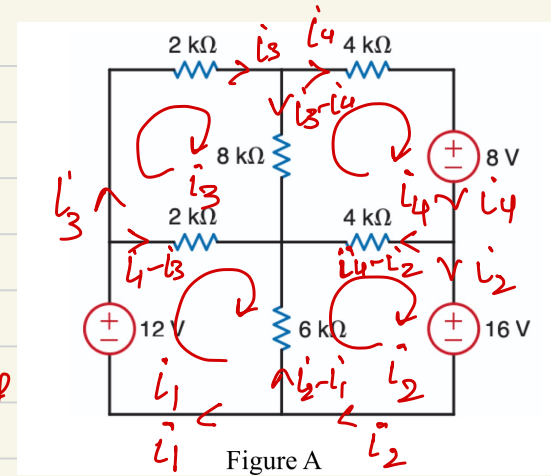
$$\text{for } V_4: (B) - 2(A) \Rightarrow V_4 - V_2 - 2V_2 + 2V_1 = 0 \Rightarrow V_4 = 3V_2 - 2V_1 \\ = -3.707 \text{ V}$$

Q3. A. Power dissipated in $6k\Omega$ resistor.

Nodes: 6

Branches: 9, Meshes: $9 - 6 + 1 = 4$

It is a planar circuit, mesh analysis can be used, and presence of voltage sources makes applying KVL a preferred choice.



mesh 1 :-

$$12 - 2(i_1 - i_3) + (i_2 - i_1)6 = 0$$

, "i" are in mA.

$$\Rightarrow -8i_1 + 6i_2 + 2i_3 + 12 = 0 \quad \text{--- (1)}$$

mesh 2

$$-6(i_2 - i_1) + 4(i_4 - i_2) - 16 = 0$$

$$\Rightarrow 6i_1 - 10i_2 + 4i_4 - 16 = 0 \quad \text{--- (2)}$$

mesh 3 :-

$$-2i_3 - 8(i_3 - i_4) + 2(i_1 - i_3) = 0$$

$$2i_1 - 12i_3 + 8i_4 = 0 \quad \text{--- (3)}$$

mesh 4 :-

$$8(i_3 - i_4) - 4i_4 - 8 - 4(i_4 - i_2) = 0$$

$$4i_2 + 8i_3 - 16i_4 - 8 = 0 \quad \text{--- (4)}$$

$\Rightarrow$ Four variables and four equations to solve for it.

If someone looks at the circuit *prima facie*, although it will have 5 nodal equations, three node voltages are known:

$V_3 = 16V$, $V_4 = 24V$, $V_1 = 12V$
 It leaves V_2 & V_5 to be calculated $\rightarrow$

Applying KCL at V_2 :-

$$\frac{V_2 - V_1}{2} + \frac{V_2}{6} + \frac{V_2 - V_3}{4} + \frac{V_2 - V_5}{8} = 0$$

using values :-

$$V_2 \left[\frac{1}{2} + \frac{1}{6} + \frac{1}{4} + \frac{1}{8} \right] - 6 - 4 - \frac{V_5}{8} = 0$$

$$\Rightarrow \frac{25}{24} V_2 - \frac{V_5}{8} - 10 = 0 \quad \text{--- (1)}$$

Applying KCL at V_5 :-

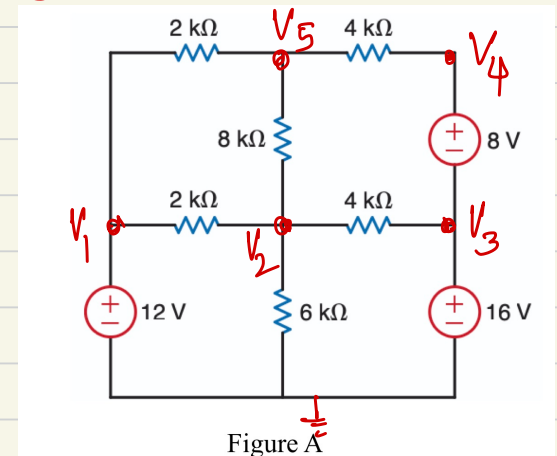
$$\frac{V_5 - V_1}{2} + \frac{V_5 - V_4}{4} + \frac{V_5 - V_2}{8} = 0$$

$$V_5 \left[\frac{1}{2} + \frac{1}{4} + \frac{1}{8} \right] - 6 - 6 - \frac{V_2}{8} = 0$$

$$\frac{7}{8} V_5 - \frac{V_2}{8} - 12 = 0 \quad \text{--- (2)}$$

$$7 \times (1) + (2) \Rightarrow \left(\frac{7 \times 25}{24} - \frac{1}{8} \right) V_2 - 70 - 12 = 0 \Rightarrow \frac{25 \times 7 - 3}{24} V_2 = 82$$

$$\Rightarrow V_2 = 11.44 V$$



$$\text{Power} = \frac{V_2^2}{6K} = \frac{(11.44)^2}{6K} = 0.022 \text{ W}$$

→ Learning: although the number of nodal/mesh equations could guide us to use a preferred method, always look at number of known & unknown variables available for each method.

Q.3B

Nodes: 8, Branches: 11

Mesh: 4

Mesh analysis appears to be the preferred choice unless some node voltages (out of 7) are already known.
↳ only 1

Let us go ahead with the mesh analysis:-

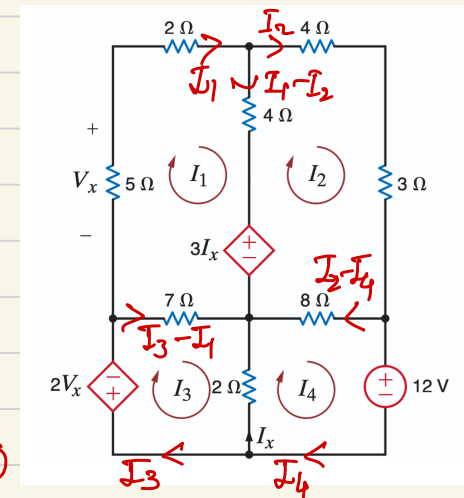
Mesh 1:- $-5I_1 - 2I_1 - 4(I_1 - I_2) - 3I_x + 7(I_3 - I_1) = 0$

(A) → $-18I_1 + 4I_2 + 10I_3 - 3I_4 = 0$ — (1)

Mesh 2:-

$3I_x + 4(I_1 - I_2) - 7I_2 - 8(I_2 - I_4) = 0$

(A) → $4I_1 - 19I_2 - 3I_3 + 11I_4 = 0$ — (2)



$I_x = I_4 - I_3$ — (A)

$V_x = -5I_1$ — (B)

Mesh 3:-

$$\textcircled{B} \rightarrow -2V_x - (I_3 - I_1)7 + (I_4 - I_3)2 = 0$$
$$17I_1 - 9I_3 + 2I_4 = 0 \quad \text{---} \textcircled{3}$$

Mesh 4:-

$$-(I_4 - I_3)2 + 8(I_2 - I_4) - 12 = 0$$
$$8I_2 + 2I_3 - 10I_4 - 12 = 0 \quad \text{---} \textcircled{4}$$

4 linear equations, 4 variables $\rightarrow$

$$\begin{bmatrix} -18 & 4 & 10 & -3 \\ 4 & -19 & -3 & 11 \\ 17 & 0 & -9 & 2 \\ 0 & 8 & 2 & -10 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 12 \end{bmatrix}$$

$$[R][I] = [V]$$

$$I = [\text{inv}(R)][V]$$

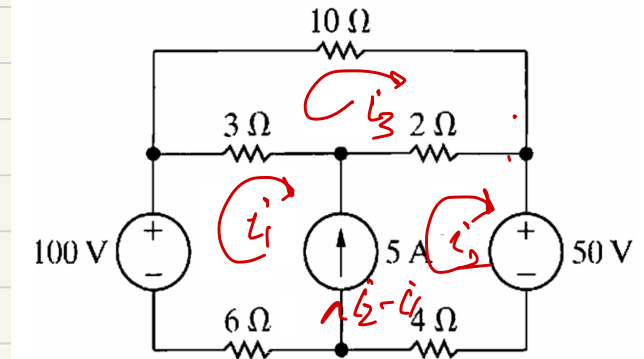
Q4

Nodes : 6, Branches : 8, Mesher : 3

known node voltages :- 1, 4 unknown node voltages

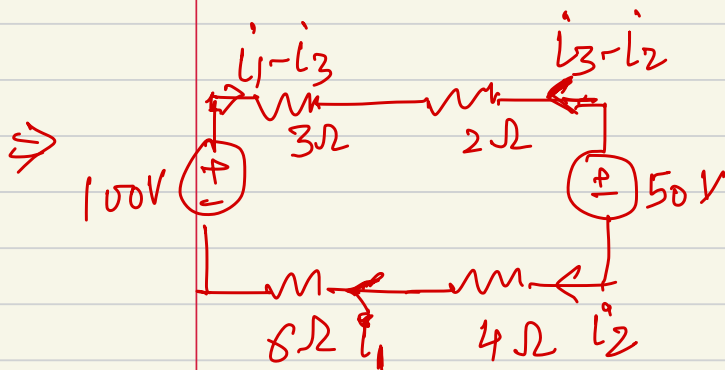
Mesh Analysis :

Problem faced $\rightarrow$ do not know the voltage across the current source.



$$i_2 - i_1 = 5A \quad \text{--- (A)}$$

Concept of Supermesh :- Merging KVL of mesh 1 & 2, which is similar to omitting the voltage drop across 5A source after writing the mesh 1 & mesh 2 eqns independently.



$$\Rightarrow 100 - 3(i_1 - i_3) + 2(i_3 - i_2) - 50 - 4i_2 - 6i_1 = 0$$

$$- 9i_1 - 6i_2 + 5i_3 + 50 = 0 \quad \text{--- (1)}$$

mesh 3 :- $10i_3 - 2(i_3 - i_2) + 3(i_1 - i_3) = 0$

$$\Rightarrow 3i_1 + 2i_2 - 15i_3 = 0 \quad \text{--- (2)}$$

$$\textcircled{A} \rightarrow \textcircled{1} \rightarrow -9i_1 - 6(5+i_1) + 5i_3 + 50 = 0$$

$$\Rightarrow -15i_1 + 5i_3 + 20 = 0 \quad \text{--- } \textcircled{3}$$

$$\textcircled{A} \rightarrow \textcircled{2} \rightarrow 3i_1 + 2(5+i_1) - 15i_3 = 0$$

$$5i_1 - 15i_3 + 10 = 0 \quad \text{--- } \textcircled{4}$$

$$\textcircled{3} + 3 \times \textcircled{4} \Rightarrow -40i_3 + 50 = 0 \Rightarrow i_3 = \frac{5}{4} = 1.25 \text{ A}$$

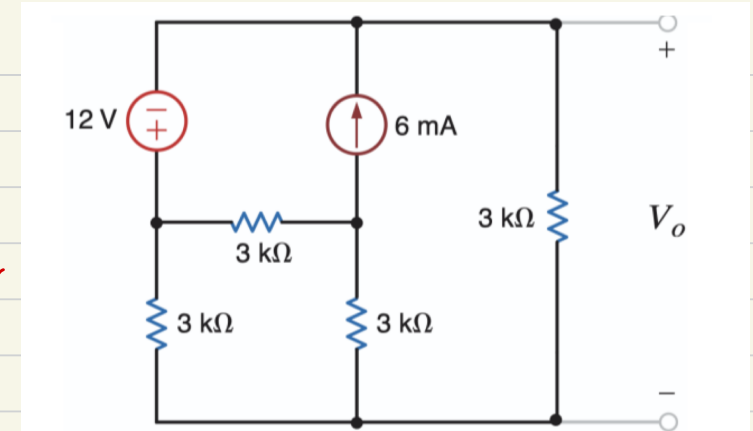
$$i_1 = -2 + 3 \times 1.25 = 1.75 \text{ A}$$

current through 3Ω resistor, $i_1 - i_3 = 0.5 \text{ A}$.

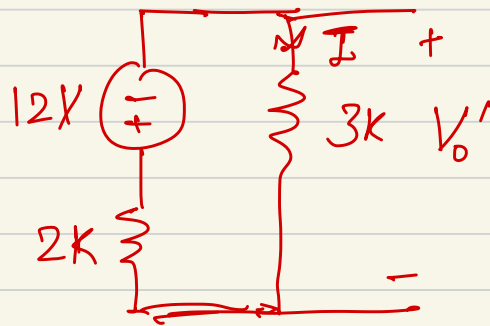
power dissipation, $\Rightarrow 0.5^2 \times 3 = 0.75 \text{ W}$

Q.5

As both are independent sources and the given circuit is a linear circuit, we can use superposition theorem to calculate the combined V_o .



When current source is open circuit & voltage source is active :-



$\Leftrightarrow$



$$-5I + 12 = 0, I = -\frac{12}{5} \text{ mA} \Rightarrow V_o' = -\frac{36}{5} \text{ V}$$

Second case:- when current source is active and voltage source is shorted.

$$V_0'' = 3I$$

Bigger loop:- (blue)

$$-3(6 - I - I_1) + 3I = 0 \quad \text{--- (1)}$$

Smaller loop- (black)

$$-3(6 - I - I_1) - 3(6 - I_1) + 3I_1 = 0 \quad \text{--- (2)}$$

$$(1) \Rightarrow -18 + 6I + 3I_1 = 0 \Rightarrow I_1 + 2I = 6 \quad \text{--- (A)}$$

$$(2) \Rightarrow -18 + 3I + 9I_1 - 18 = 0 \Rightarrow 3I_1 + I = 12 \quad \text{--- (B)}$$

$$3 \times (A) - (B) \Rightarrow 5I = 6 \Rightarrow I = \frac{6}{5} \text{ mA} \Rightarrow V_0'' = \frac{18}{5} \text{ V}$$

Using superposition theorem, $V_0 = V_0' + V_0'' = -\frac{18}{5} \text{ V}$

